# Agent-Based SIR Model in AnyLogic

In this exercise we build an epidemic as an agent-based model (ABM) in AnyLogic: 1000 individual people who catch, spread and recover from a disease.

## Install AnyLogic PLE (free)

The Personal Learning Edition is free and needs no licence key. The current release at the time of writing is AnyLogic 8.9.10. Install it before the exercise: the download and setup take 10–20 minutes on a normal connection.

### Check your computer

| | Requirement |
|---|---|
| Windows | Windows 10 or 11, 64-bit |
| macOS | macOS 12 to 15, Intel or Apple silicon (M1/M2/M3/M4) |
| Linux | Ubuntu 18, 20 or 22 (64-bit); other distributions may work |
| Disk space | 1.5 GB free |
| Memory | 4–8 GB recommended |

### Step 0.1. Download

1. Go to anylogic.com/downloads.
2. Find **Personal Learning Edition** and click **Download**.
3. Fill in the short form and submit it. Use an email address you can check now.
4. The download should start, or you will be offered a file for your operating system. If nothing starts, check your email (and spam folder) for a download link.

On macOS, pick the file that matches your Mac: **ARM** for Apple silicon, **x64** for Intel. To check, open the Apple menu → **About This Mac**.

### Step 0.2. Install

- **Windows:** Run the downloaded `.exe` file. Accept the licence agreement and keep the default folder. Click **Install**, then **Finish**.

- **macOS:** Open the downloaded `.dmg` file and drag **AnyLogic** into **Applications**. On first launch, if macOS warns that the app was downloaded from the internet, click **Open**.
- **Linux:** Make the downloaded `.tgz.bin` file executable (`chmod +x <file>`), then run it and follow the prompts.

On Windows you can also install from a terminal with `winget install AnyLogic.AnyLogic.Personal`.

### Step 0.3. First launch

1. Start AnyLogic. The PLE activates itself; there is no key to type. If an activation window appears, choose the Personal Learning Edition option and follow the prompts.
2. A **Welcome** page opens with example models and tutorials. Close it; you can reopen it later from the **Help** menu.
3. Check the version under **Help → About AnyLogic**. It should say 8.9 or later.

#### The AnyLogic window

- **Projects** panel (top left): the tree of your model. Double-click `Main` or `Person` here to open that agent's diagram.
- **Palette** panel (left, below Projects): the toolbox. Each section (Agent, Statechart, Presentation, Analysis) holds elements you drag onto the diagram.
- **Graphical editor** (centre): the canvas where you place elements.
- **Properties** panel (right): settings of whatever you last clicked. Most of your typing happens here.
- **Problems** panel (bottom): errors. If the model will not run, look here first. Double-click an error to jump to it.
- **Run** button (green triangle in the toolbar): builds and runs the model.

The PLE limits model size (for example, at most 10 agent types per model). This lab uses 2 agent types, so the limits do not affect it. The PLE licence allows classroom and personal learning use only.

## Build the agent-based SIR model

#### The SIR Model

Every person is in exactly one of three states. **S**usceptible (**S**) people can catch the disease. **I**nfected (**I**) people are sick and can pass it on. **R**ecovered (**R**) people are immune and play no further part. People move only one way: **S → I → R**.

The logic of every person is simple. While a person is infected, they make random contacts with other people at a rate of 5 per day. Each contact passes on the disease with probability 0.05, but only if the other person is still susceptible. After about 15 days, on average, the infected person recovers.

With these values the basic reproduction number is R0 = contactRate × infectivity × illnessDuration = 5 × 0.05 × 15 = 3.75. Each early case infects about 3.75 others, so an epidemic takes off.

**Terminology from Agent-Based Modeling**

- **Agent**: one individual in the model. Here, one person.
- **Agent type**: the template every agent is made from. We will call ours `Person`.
- **Population**: a collection of many agents of one type. Ours is called `people`.
- **Main**: the top-level agent that AnyLogic creates for you. It holds the population, the parameters and the charts.
- **Statechart**: a diagram of the states an agent can be in and the rules (transitions) for moving between them.
- **Transition**: an arrow between two states. It fires when its trigger happens (a rate, a timeout, or a message).
- **Message**: a signal one agent sends to another. We use it to pass on the infection.
- **Parameter**: a fixed input value you can change between runs.

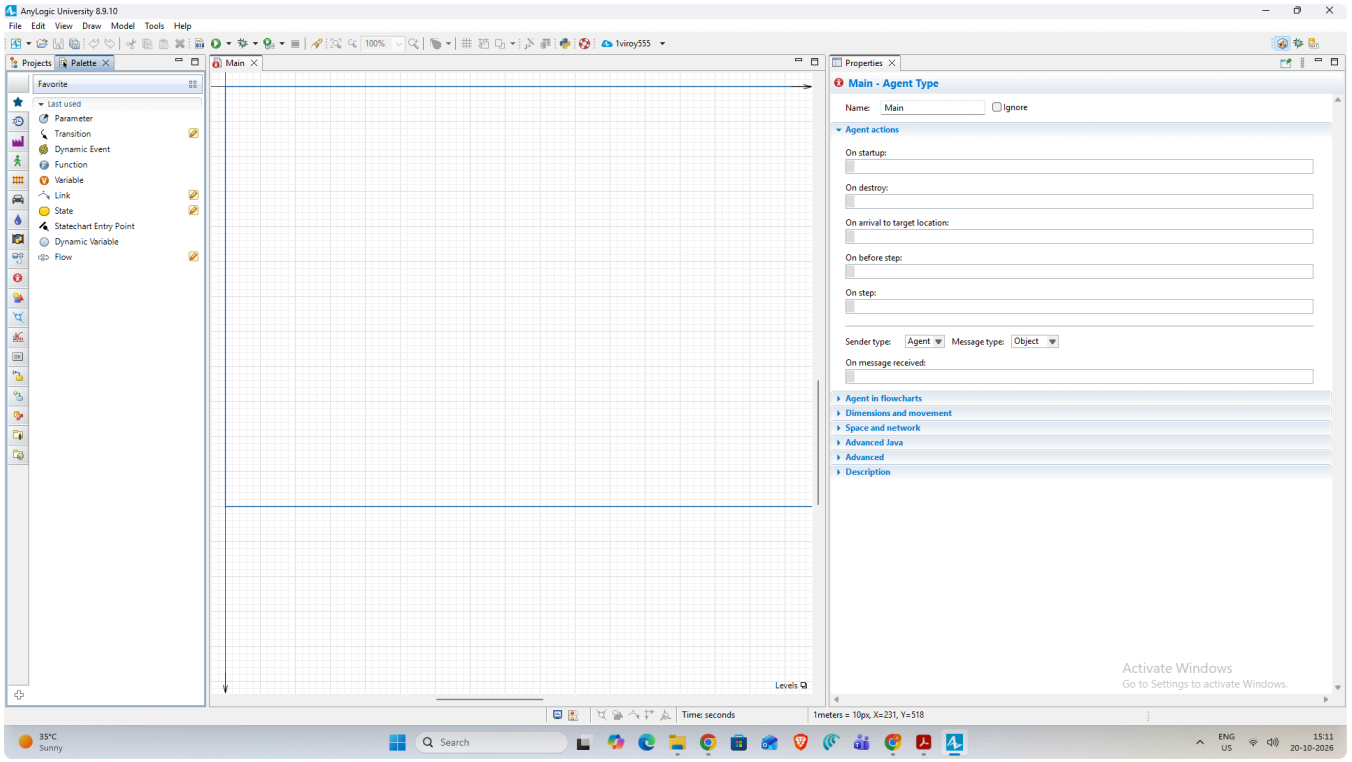
## Parameter values used throughout

| Parameter name | Value | Meaning |
|---|---|---|
| `N` | 1000 | Number of people |
| `contactRate` | 5 | Contacts each infected person makes per day |
| `infectivity` | 0.05 | Probability that one contact with a susceptible person passes on the disease |
| `illnessDuration` | 15 | Average number of days a person stays infected |
| `initialInfected` | 5 | People infected at the start |

## Step 1. Create a new model

1. Open AnyLogic. Choose **File → New → Model**.
2. Model name: `SIR_Lab`. Pick a folder you can find again.
3. Set **Model time units** to **days**. This matters: every rate you type later is per day.

4. Click **Finish**. AnyLogic opens an empty diagram called `Main`.



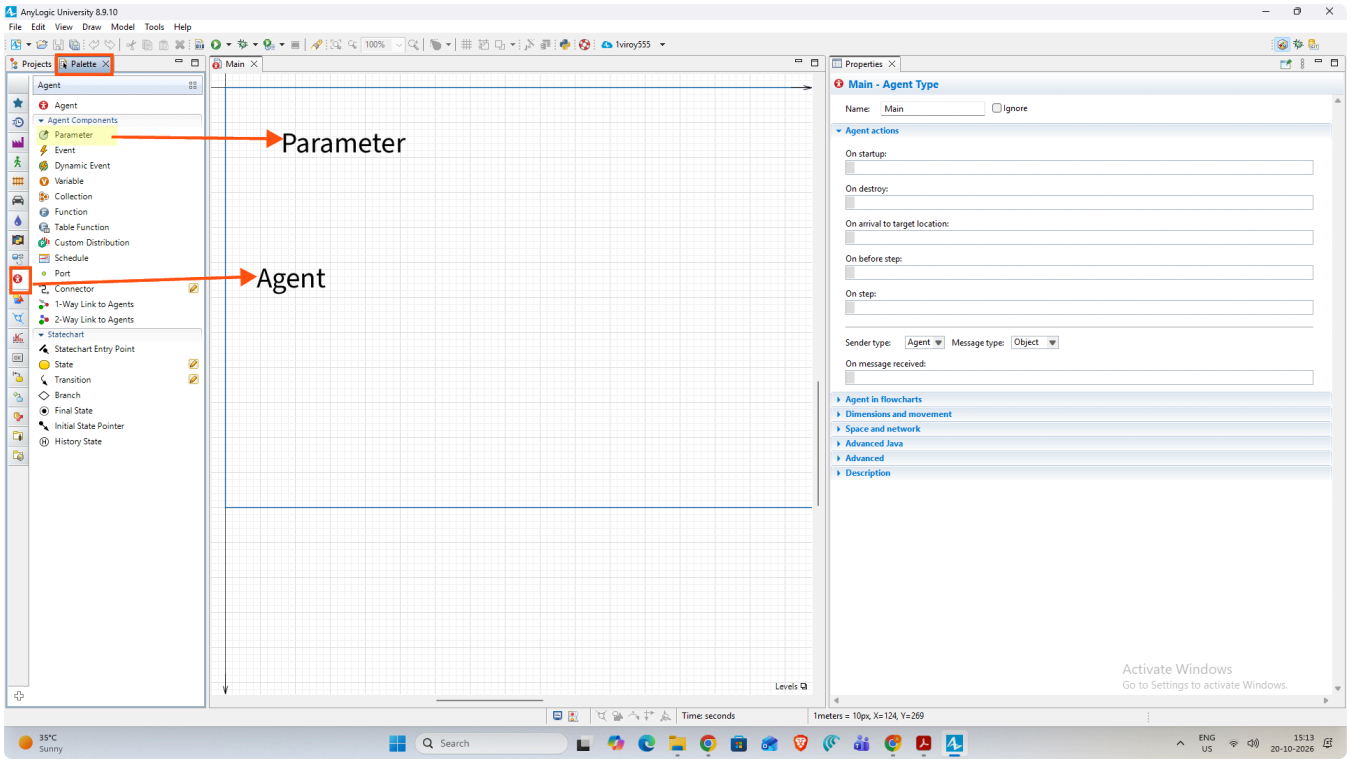

## Step 2. Add the parameters to Main

Parameters hold the input numbers, so you can change them in one place.

1. In the **Palette** panel, open the **Agent** section. Drag a **Parameter** onto the `Main` canvas.
2. In **Properties** (right), set Name = `N`, Type = `int`, Default value = `1000`.
3. Repeat for the other four parameters:

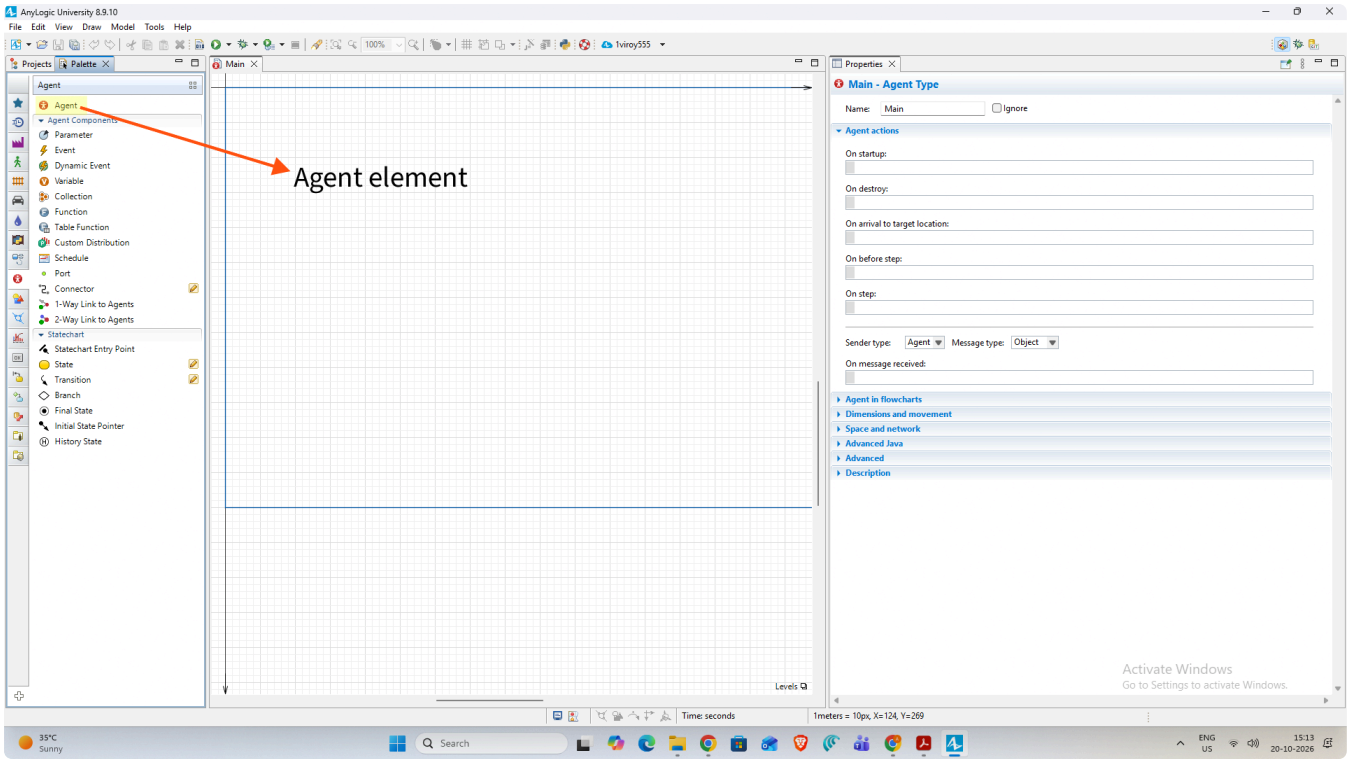
| Name | Type | Default value |
|---|---|---|
| `contactRate` | double | 5 |
| `infectivity` | double | 0.05 |
| `illnessDuration` | double | 15 |
| `initialInfected` | int | 5 |

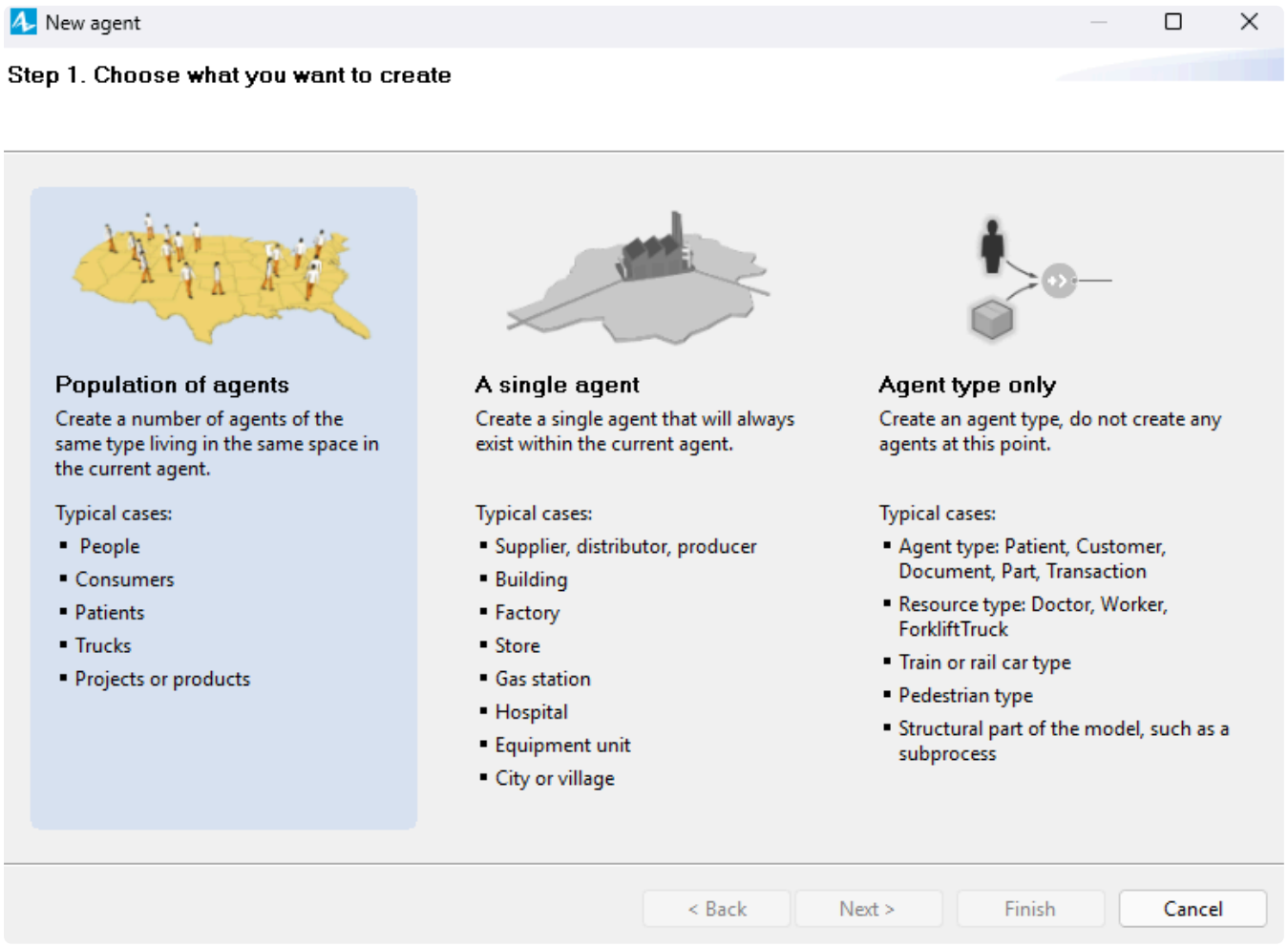
Names are case-sensitive. `contactRate` and `ContactRate` are different things to AnyLogic. Type them exactly as shown.

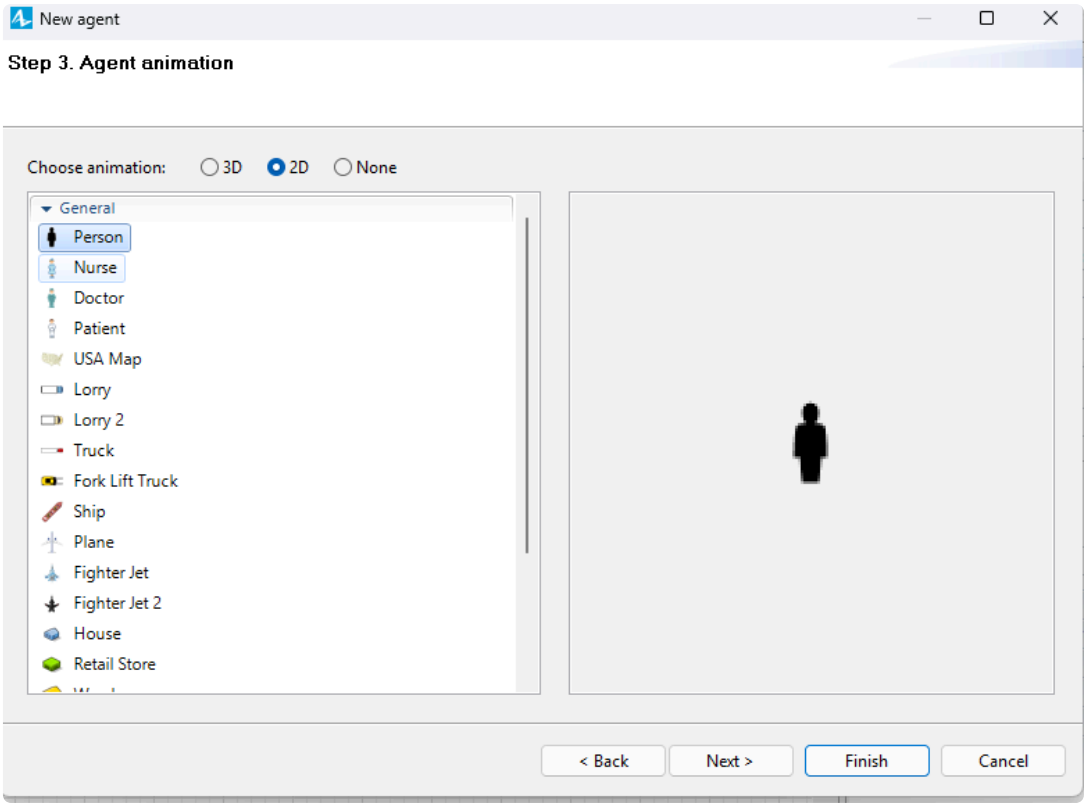



## Step 3. Create the population of people

1. In **Palette → Agent**, drag the **Agent** element onto `Main`. A wizard opens.
2. Choose **Population of agents**, then **I want to create a new agent type**. Click **Next**.
3. Agent type name: `Person`. The population name fills in as `people`. Click **Next**.
4. Animation: choose **2D** and any person icon. Click **Next**.
5. Agent parameters: add none. Click **Next**.
6. Population size: type `100`. Click **Next**.
7. Click the new `people` element on `Main`. In Properties, change **Initial number of agents** from `100` to `N`. Now the population follows the parameter.


Agent element


New agent
Step 1. Choose what you want to create
Population of agents
Create a number of agents of the same type living in the same space in the current agent.
Typical cases:
People
Consumers
Patients
Trucks
Projects or products
A single agent
Create a single agent that will always exist within the current agent.
Typical cases:
Supplier, distributor, producer
Building
Factory
Store
Gas station
Hospital
Equipment unit
City or village
Agent type only
Create an agent type, do not create any agents at this point.
Typical cases:
Agent type: Patient, Customer, Document, Part, Transaction
Resource type: Doctor, Worker, ForkliftTruck
Train or rail car type
Pedestrian type
Structural part of the model, such as a subprocess
< Back
Next >
Finish
Cancel




## Step 4. Draw the statechart inside Person

The statechart is the heart of the model. It tells each person which state they are in and when they move on.

1. In the **Projects** panel, double-click `Person` to open its diagram. You will see the person icon near a small blue cross (the origin).
2. In **Palette → Agent → Statechart**, drag a **Statechart Entry Point** onto the canvas, a little to the side of the icon.
3. Drag three **State** elements onto the canvas, one below the other. Make each state about twice as tall as the default; you will need room inside `Infected` later.
4. Click each state and set its **Name** and **Fill color** in Properties:
   - `Susceptible`, green
   - `Infected`, red
   - `Recovered`, grey
5. Drag the arrowhead of the entry point onto the top edge of `Susceptible`, until it snaps. Every person now starts in `Susceptible`.

## Step 5. Add the transitions

To draw a transition, double-click **Transition** in **Palette → Agent → Statechart**. This turns on drawing mode. Click once on the edge of the start state, then click on the edge of the end state. Then select the transition and fill in its properties.

### Transition 1: Susceptible → Infected (getting infected)

- Name: `infection`
- Triggered by: **Message**
- Message type: **String**
- Fire transition: **On particular message**
- Message: `"Infection"` (with the double quotes)

This means: a susceptible person becomes infected the moment they receive the message "Infection". A person who is already infected or recovered ignores the message, because only `Susceptible` has this arrow.

### Transition 2: inside Infected (making contacts)

This arrow starts and ends on the edge of `Infected`, making a small loop inside the state. Double-click **Transition**, click on the left edge of `Infected`, click once or twice inside the state to bend the line, then click on the right edge of `Infected`.

- Name: `contact`
- Triggered by: **Rate**
- Rate: `main.contactRate`, units **per day**
- Action:

```
if (randomTrue(main.infectivity)) {
    send("Infection", RANDOM);
}
```

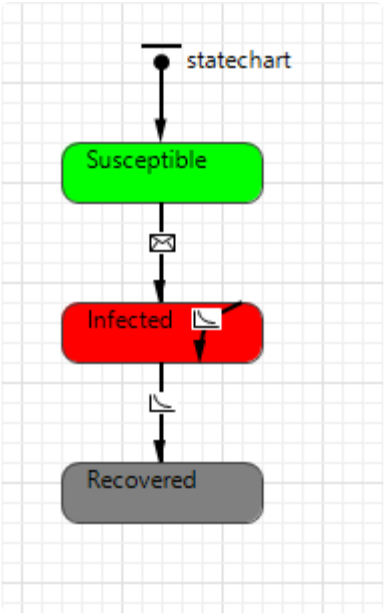
Read it like this. About 5 times a day, at random moments, the infected person makes a contact. With probability 0.05 the contact transmits, and the message "Infection" goes to one person chosen at random from the whole population. If that person happens to be susceptible, they become infected. The person stays in `Infected` after the loop fires.

`main` is how a person reaches the parameters stored on `Main`. `RANDOM` means any random member of the same population.

### Transition 3: Infected → Recovered (recovering)

- Name: `recovery`
- Triggered by: **Rate**
- Rate: `1.0 / main.illnessDuration`, units **per day**

A rate of 1/15 per day means the average stay in `Infected` is 15 days. Some people recover after 2 days and some after 40, but the mean is 15. Write `1.0` and not `1`; in Java, `1 / 15` is integer division and gives 0.

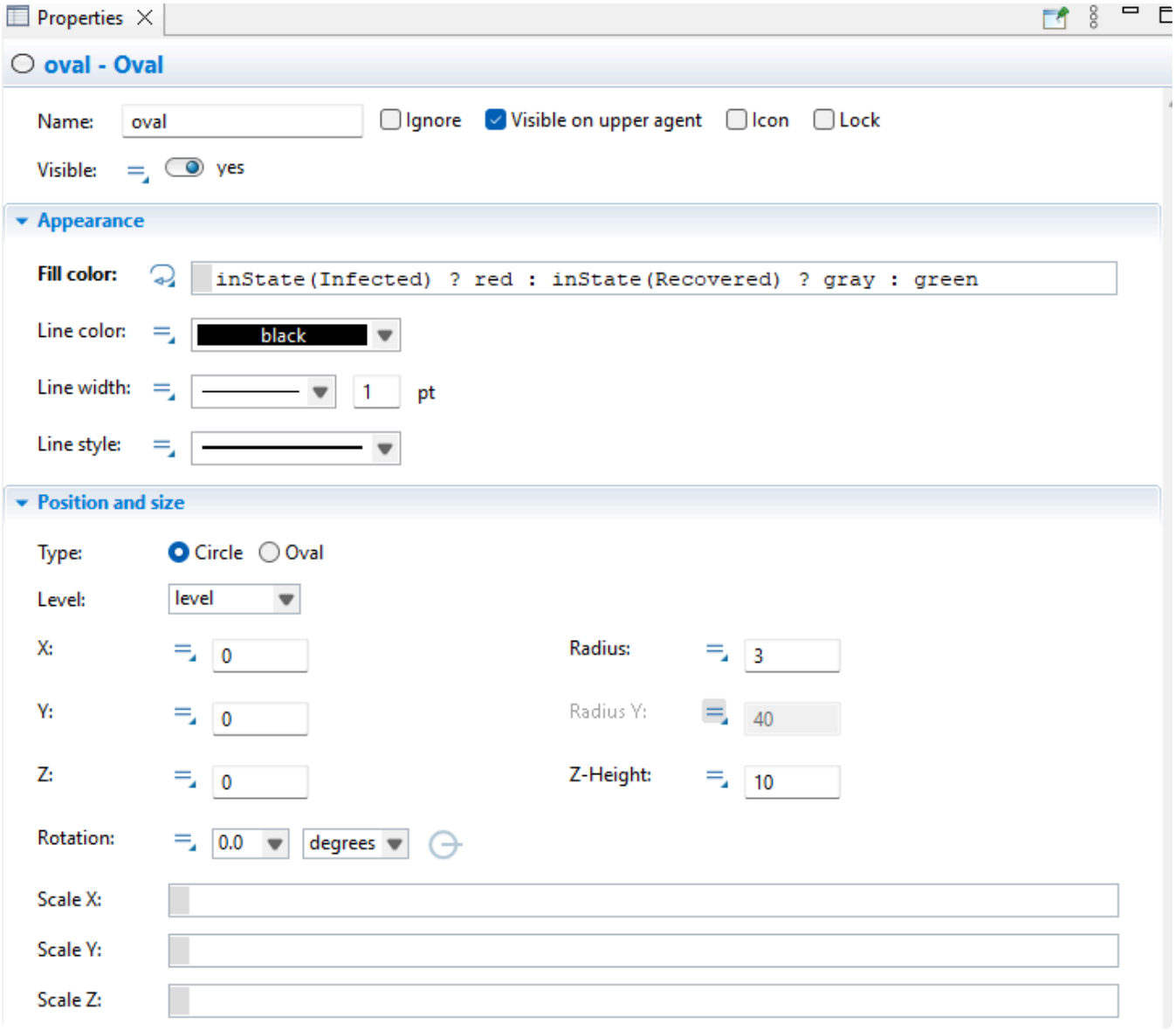



## Step 6. Colour the people by state

1. In the `Person` diagram, delete the person icon from the wizard.
2. From **Palette → Presentation**, drag an **Oval** and place its centre exactly on the blue origin cross. Set radius to `3`.
3. In the oval's properties, find **Fill color**. Click the small icon beside it to switch it to a dynamic value, and type:

```
inState(Infected) ? red : inState(Recovered) ? gray : green
```

Each person is now drawn green, red or grey according to their current state.

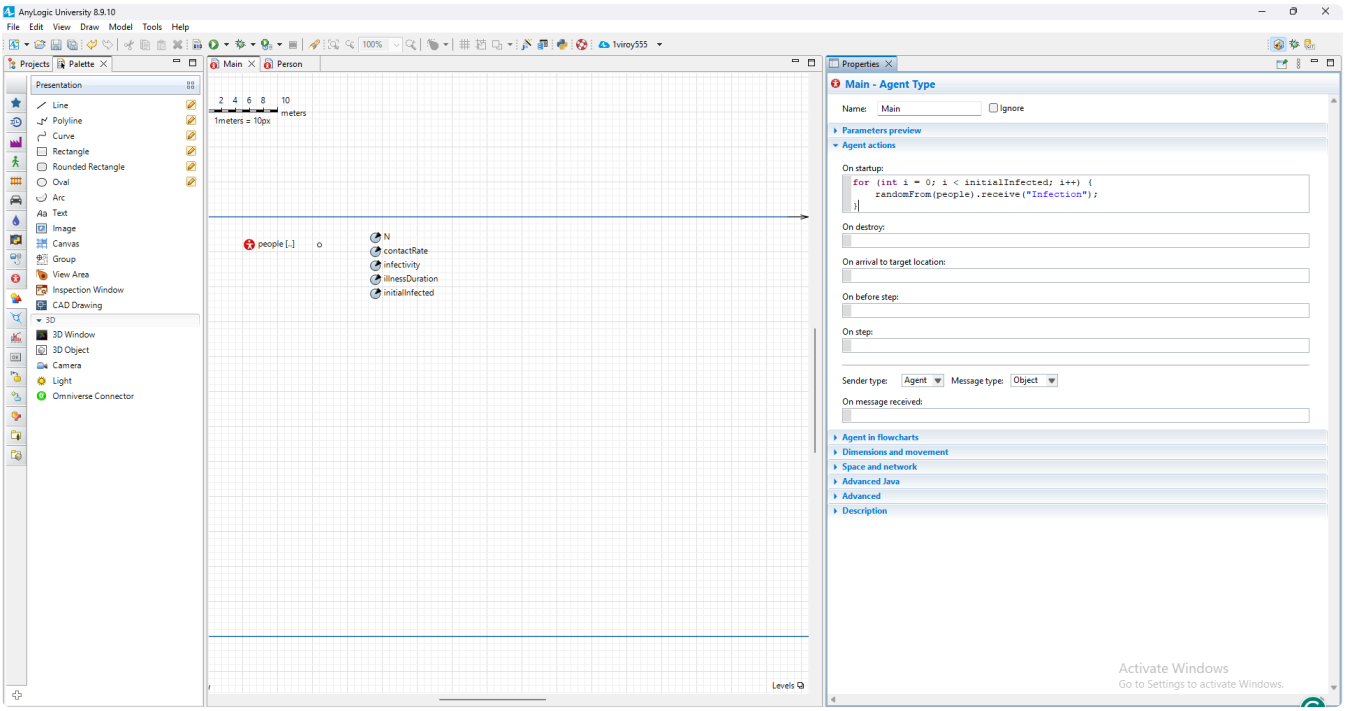



## Step 7. Infect the first few people

Go back to `Main` (double-click it in Projects). Click an empty spot on the canvas. In Properties, open **Agent actions** and type into **On startup**:

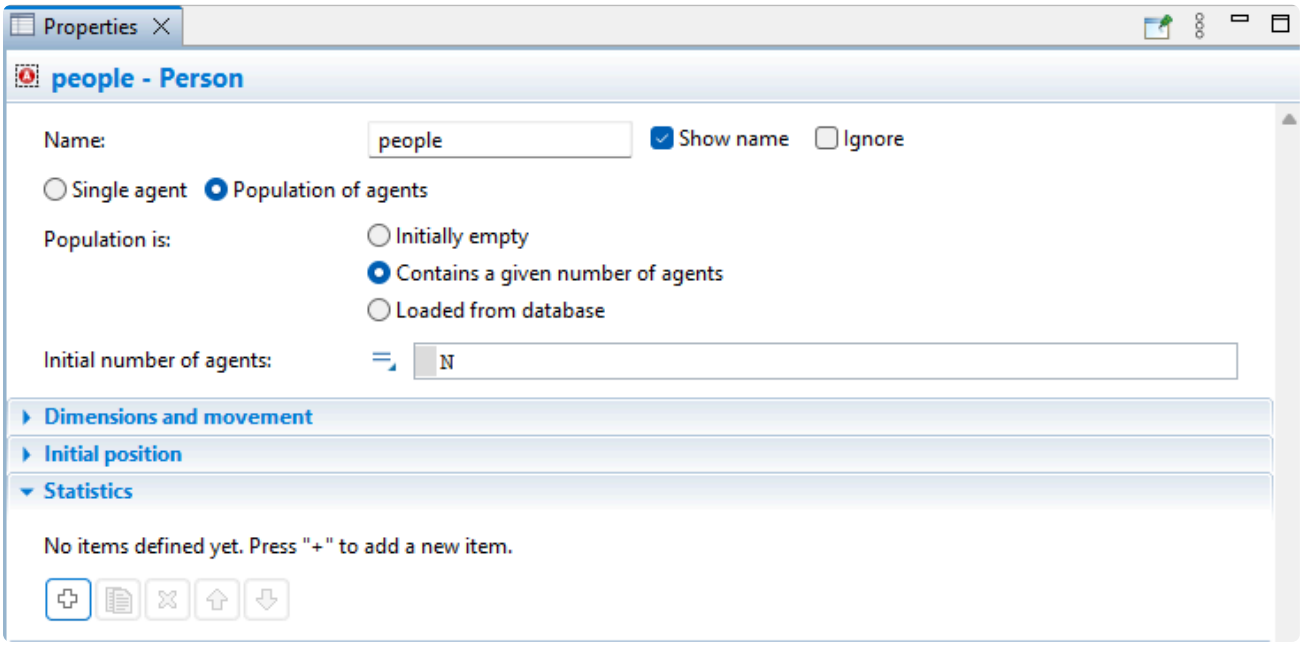
```
for (int i = 0; i < initialInfected; i++) {
    randomFrom(people).receive("Infection");
}
```

This picks 5 random people when the model starts and hands each one the "Infection" message.




## Step 8. Count people in each state

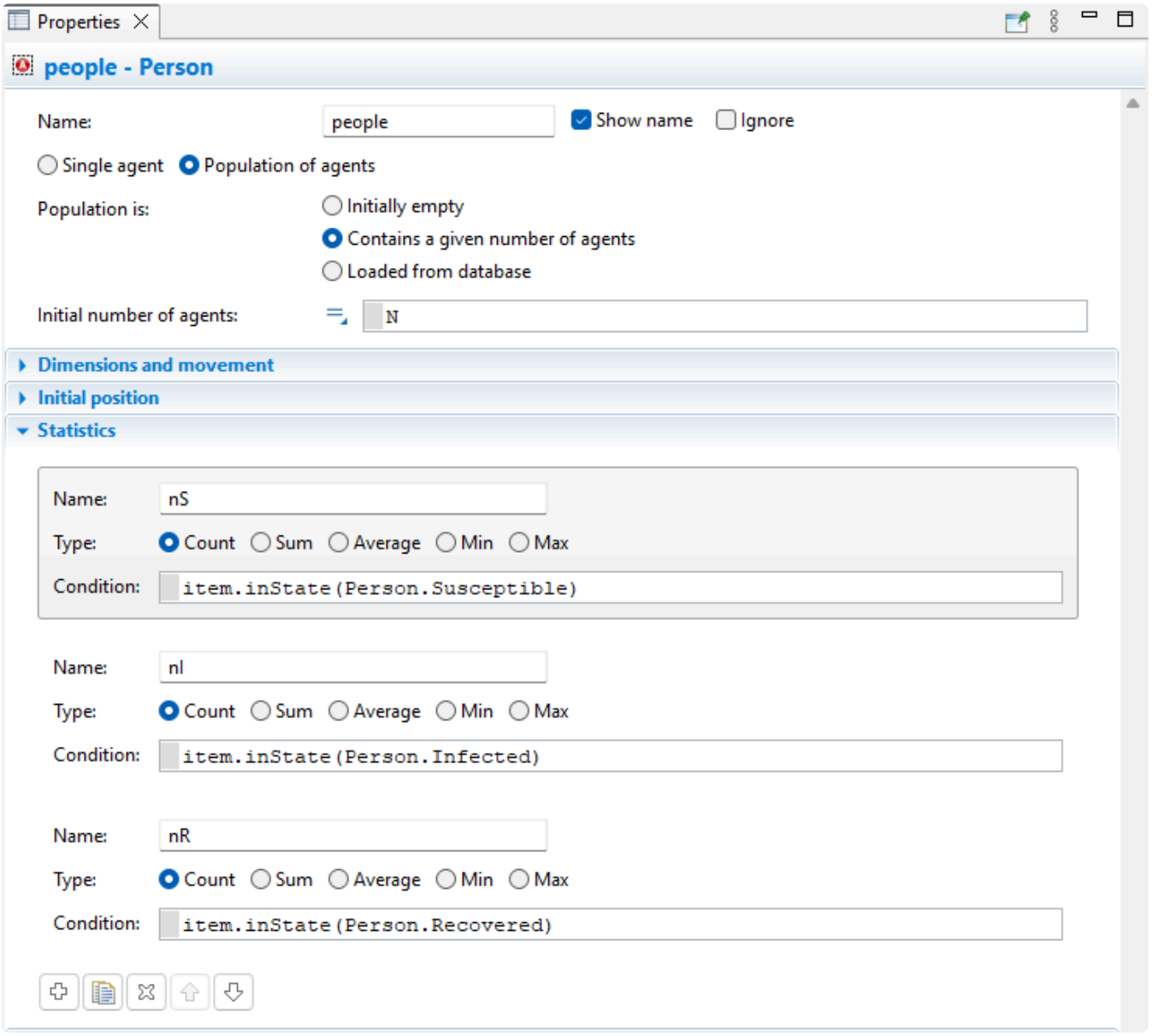
1. On `Main`, click `people`. In Properties, scroll to **Statistics** and click **+** three times.




2. Fill them in:

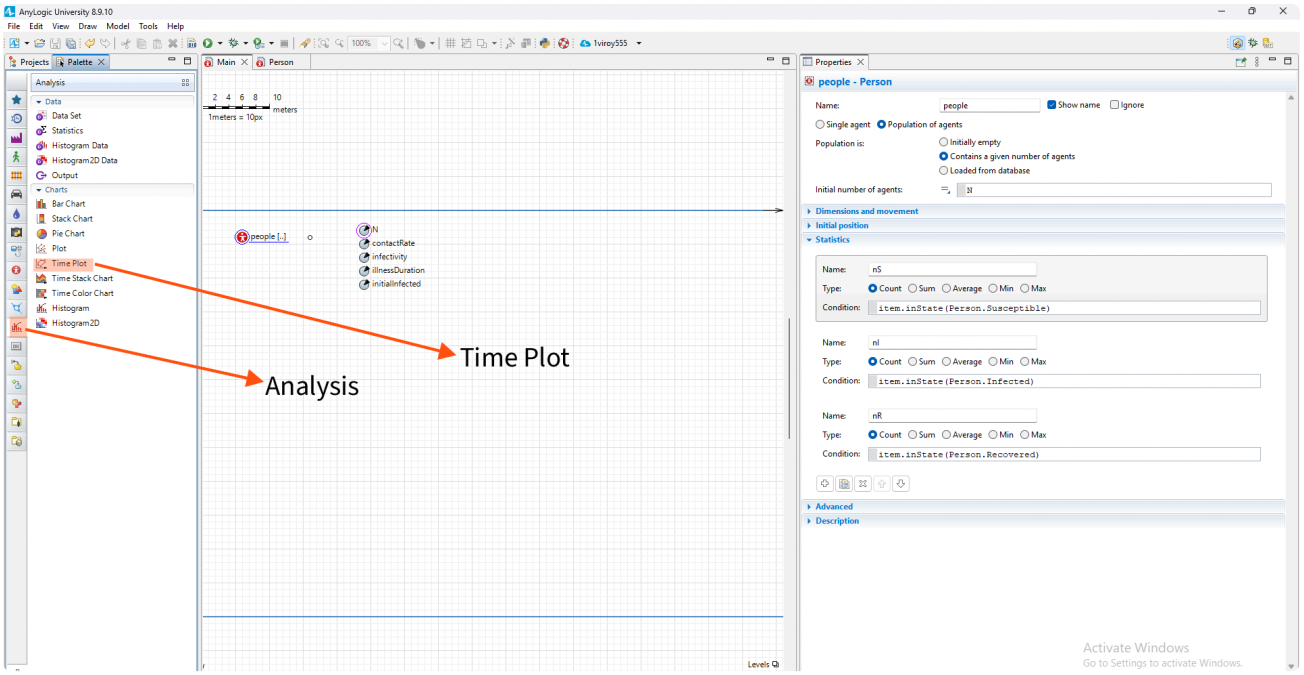
| Name | Type | Condition |
|---|---|---|
| `nS` | Count | `item.inState(Person.Susceptible)` |
| `nI` | Count | `item.inState(Person.Infected)` |
| `nR` | Count | `item.inState(Person.Recovered)` |

`item` stands for each person in the population. `nS` counts the people for whom the condition is true.




## Step 9. Add a chart

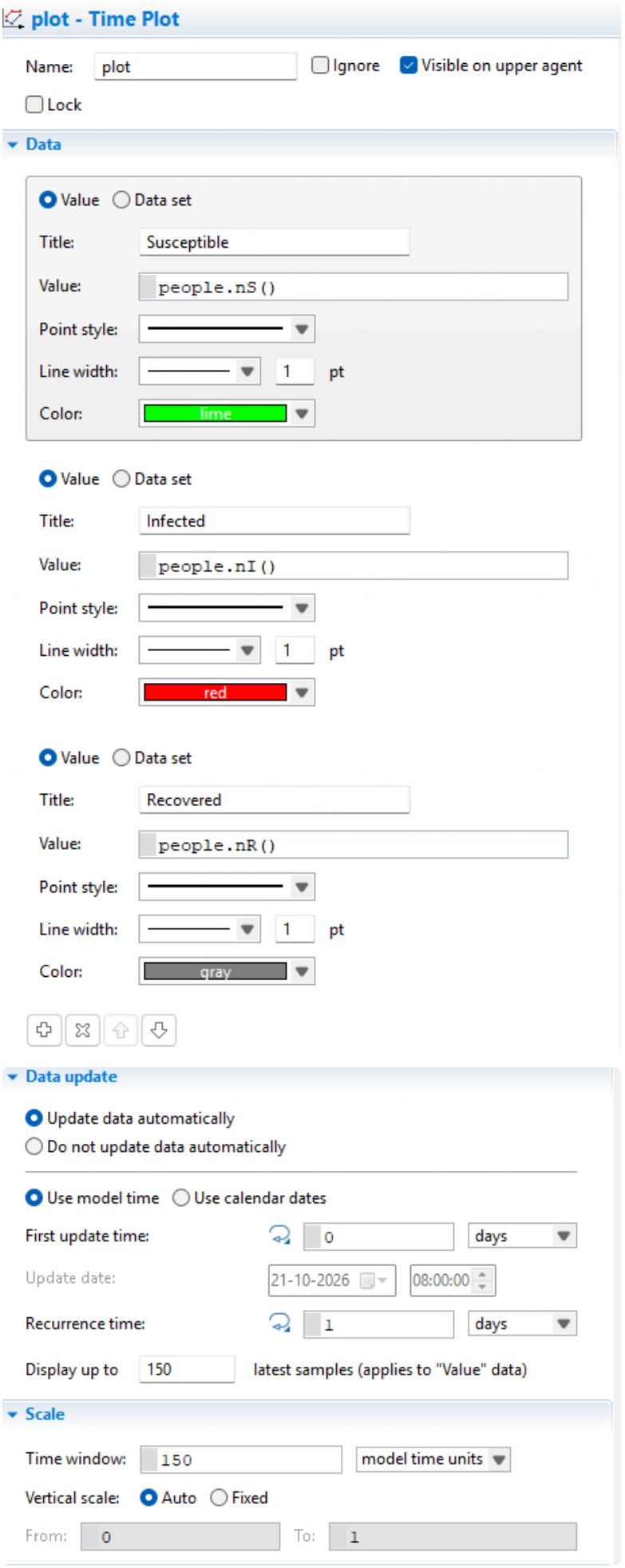
1. From **Palette → Analysis**, drag a **Time Plot** onto `Main`, beside the people area. Stretch it to a comfortable size.




2. In its Properties, under **Data**, click **Add data item** three times:

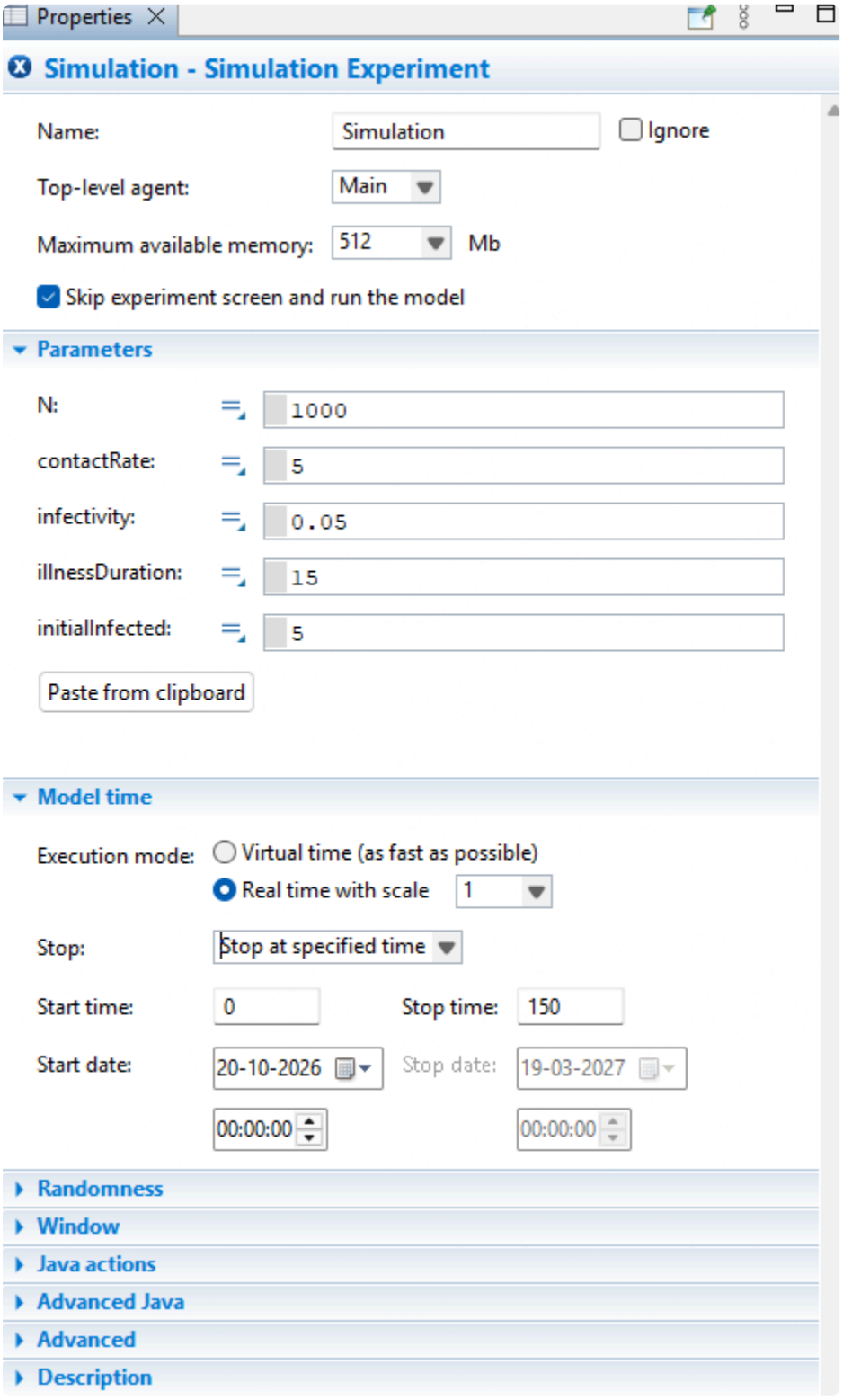
| Title | Value | Colour |
|---|---|---|
| Susceptible | `people.nS()` | green |
| Infected | `people.nI()` | red |
| Recovered | `people.nR()` | grey |

3. Set **Time window** to `150` (days).


plot - Time Plot
Name: plot
Ignore
Visible on upper agent
Lock
Data
Value
Data set
Title: Susceptible
Value: people.nS()
Point style:
Line width: 1 pt
Color: lime
Value
Data set
Title: Infected
Value: people.nI()
Point style:
Line width: 1 pt
Color: red
Value
Data set
Title: Recovered
Value: people.nR()
Point style:
Line width: 1 pt
Color: gray
Data update
Update data automatically
Do not update data automatically
Use model time
Use calendar dates
First update time: 0 days
Update date: 21-10-2026 08:00:00
Recurrence time: 1 days
Display up to 150 latest samples (applies to "Value" data)
Scale
Time window: 150 model time units
Vertical scale: Auto Fixed
From: 0
To: 1

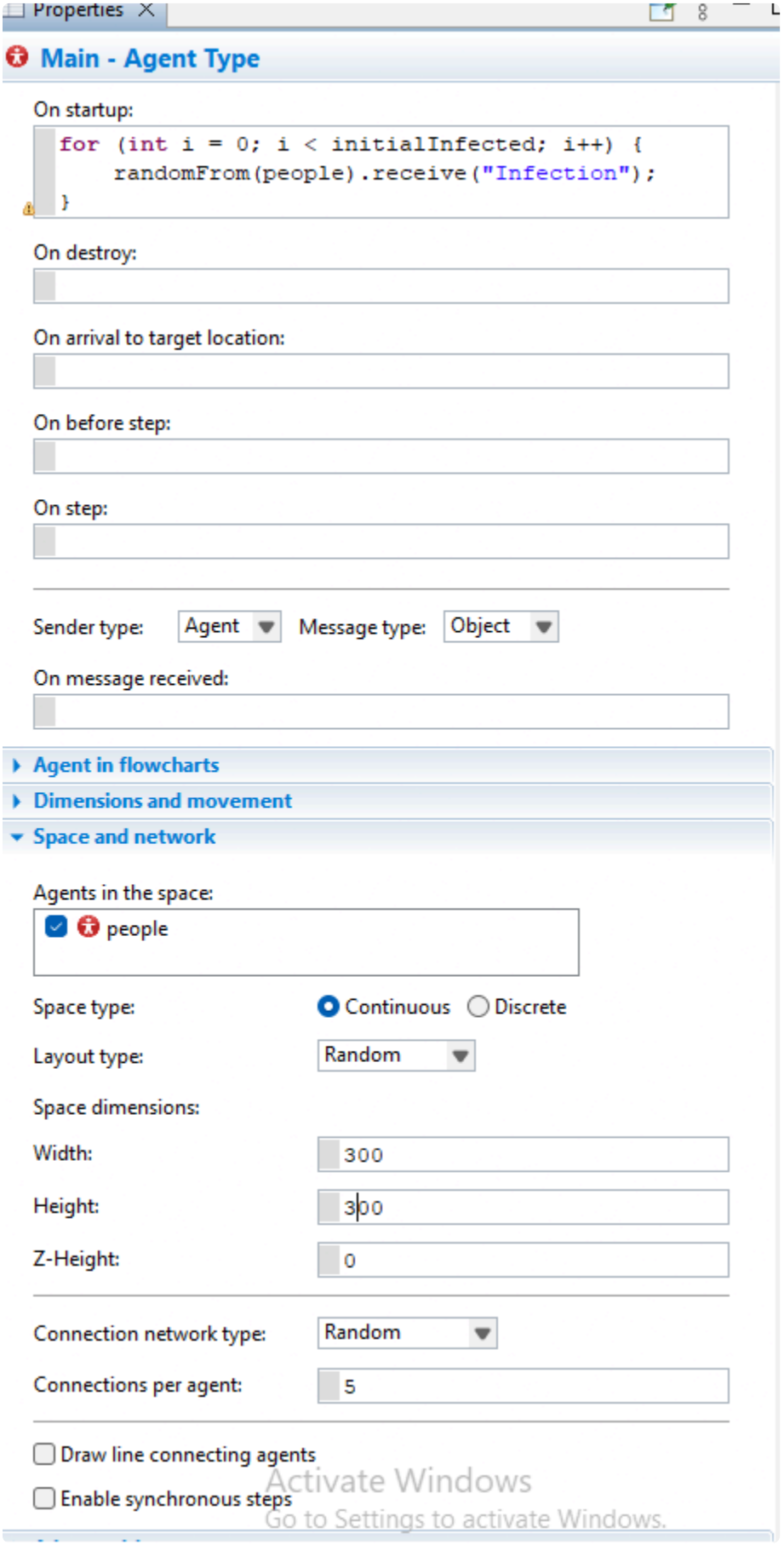
## Step 10. Set the run length and run

1. In **Projects**, click `Simulation` (under `SIR_Lab`). In Properties, under **Model time**, set **Stop** to **Stop at specified time** and **Stop time** to `150`.




2. Click the green **Run** button and choose `SIR_Lab / Simulation`. In the window that opens, click the run button.
3. Click anywhere in `Main` . Go to the properties go to the **Space and Network.** Set **Space type** to **Continuous**, **Width** to `300` and **Height** to `300`.

4. Set **Layout type** to **Random**. Set **Network type** to **User-defined** for now.



5. Speed it up with the speed control on the toolbar, or switch to virtual time mode to run as fast as possible.

### What you should see

By the end you will have 1000 people on screen, turning from green to red to grey, and a chart showing about 380 people sick at once around day 33.

- A few red dots appear, then red spreads quickly through the green.
- The red curve peaks at roughly **380–400 infected around day 35–37**.
- By day 150 almost everyone is grey: about **97% of people** catch the disease, and about 33 never do.

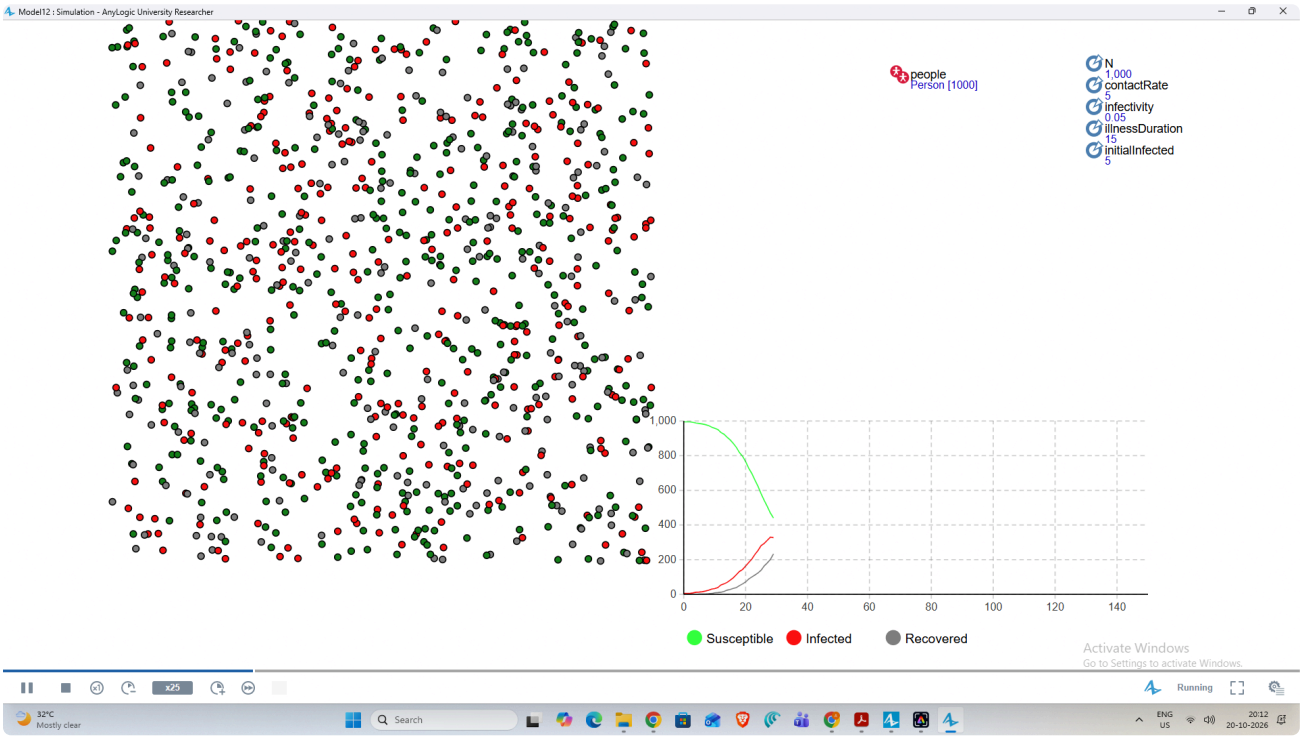



Image — Screenshot 2026-10-20 201314_edited.png

## Questions and take-home exercises

### Answer during the lab

1. Why does a recovered person ignore the "Infection" message?
2. What is R0 for `contactRate = 3`? Predict whether the epidemic will be bigger or smaller, then run it.
3. For question 3 and 4, go to the **Simulation experiments** and go to the **Randomness** and select **random seed.** Run the model 5 times. Write down the peak number infected and the day of the peak each time. How much do they vary, and why?

4. Set `initialInfected` to `1` and run 10 times. In how many runs does the disease die out early? Why can this happen?

### Take-home exercises

- [ ] **Waning immunity (SIRS).** Add a transition from `Recovered` back to `Susceptible` with rate `1.0 / 90` per day. What happens over a 2-year run?
- [ ] **Vaccination.** In **On startup**, move 30% of people straight to `Recovered` before the epidemic starts. Hint: send them a new message, `"Vaccinate"`, and add a matching transition from `Susceptible` to `Recovered`. How does the epidemic size change?
- [ ] **Different people, different contact rates.** Give `Person` its own parameter `myContactRate`, set to `uniform(1, 9)`, and use it in the `contact` transition. The average is still 5. Compare the epidemic with the original model.

## Part B: The equivalent System Dynamics model

An SD model does not track individuals. It keeps three numbers (stocks) and two rates (flows) that move people between them.

### Step B1. Add three stocks

1. Open `Main`. Move the people area and the chart aside to make room.
2. From **Palette → System Dynamics**, drag three **Stock** elements onto `Main`, left to right.
3. Name them and set **Initial value**:

| Name | Initial value |
|---|---|
| `S` | `N - initialInfected` |
| `I` | `initialInfected` |
| `R` | `0` |

The SD model starts with the same 995 susceptible and 5 infected as the ABM.

### Step B2. Add two flows

1. To draw a flow, double-click the stock `S`, then click the stock `I`. An arrow with a valve appears between them.
2. Name it `InfectionRate`. Set its formula to:

```
contactRate * infectivity * S * I / N
```

3. Double-click `I`, then click `R`. Name the new flow `RecoveryRate`. Set its formula to:

```
I / illnessDuration
```

## Step B3. Draw the links

In SD, AnyLogic wants a visible arrow (a **link**) from every element a formula uses to the element that uses it. To draw one, double-click **Link** in **Palette → System Dynamics**, click the element being used, then click the flow.

- To `InfectionRate`: links from `contactRate`, `infectivity`, `N`, `S` and `I`.
- To `RecoveryRate`: links from `illnessDuration` and `I`.

If you miss one, the **Problems** panel names the element and the flow. Draw the missing link and the error disappears.

## Step B4. Add the SD curves to the same chart

Click the Time Plot. Under **Data**, add three more data items:

| Title | Value | Colour | Line style |
|---|---|---|---|
| S (SD) | `S` | green | dashed |
| I (SD) | `I` | red | dashed |
| R (SD) | `R` | grey | dashed |

## Why the two models agree

Count new infections per day in the ABM. There are I infected people. Each makes contactRate contacts per day, and each contact transmits with probability infectivity. The person contacted is chosen at random from N people, so the chance they are susceptible is S / N. Multiply these together:

```
\text{new infections per day} = I \cdot c \cdot p \cdot \frac{S}{N} =
\frac{c\,p\,S\,I}{N}
```

This is exactly the SD formula for `InfectionRate`. Here c is `contactRate` and p is `infectivity`.

Recovery works the same way. Each infected agent recovers at rate 1 / illnessDuration, so I agents produce I / illnessDuration recoveries per day. This is the SD formula for `RecoveryRate`.

So, on average, the ABM follows the SD equations. The SD model is what the ABM looks like when you have many people, random mixing, and you only look at the totals. Two assumptions carry this result, and Part C breaks each of them:

1. The number of people involved is large enough that chance averages out.
2. Everyone is equally likely to meet everyone else ( `RANDOM` in the `send` call).

A third, hidden assumption comes from using a rate for recovery: the time spent infected is exponentially distributed.

## When this two model disagree?